

Experiment: 7

Analog Beamforming Analysis Using MATLAB

Aim

To analyze the performance of analog beamforming by studying beamforming gain, beam width, and main lobe direction using MATLAB.

Objective – 1

Analyze Beamforming Gain (dB)

Generate beamforming gain values for different antenna array sizes using MATLAB. Compare

how increasing the number of antenna elements improves the beamforming gain and enhances

signal strength.

MATLAB Code:

```
clc;
```

```
clear;
```

```
close all;
```

```
antennas = [2 4 8 16 32]; % Number of Antenna Elements
```

```
gain = 10*log10(antennas); % Beamforming Gain (dB)
```

```
figure
```

```
bar(antennas, gain)
```

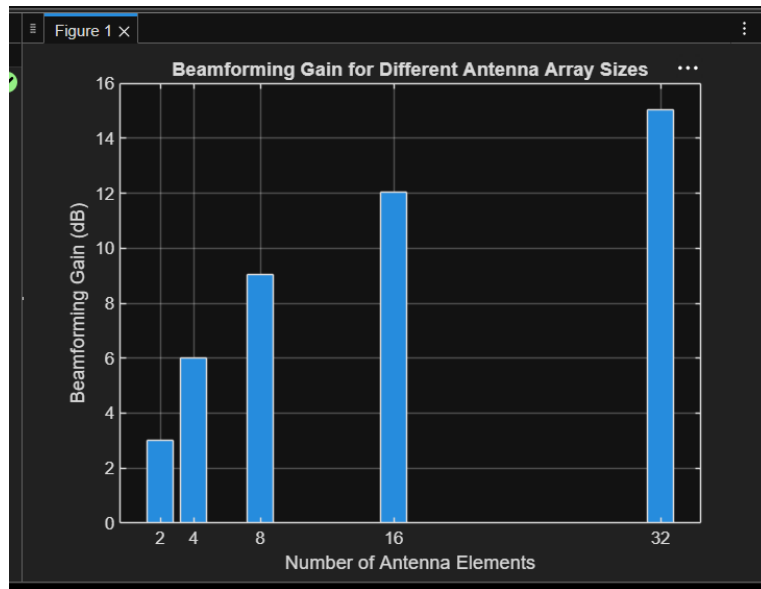
```
grid on
```

```
xlabel('Number of Antenna Elements')
```

```
ylabel('Beamforming Gain (dB)')
```

title('Beamforming Gain for Different Antenna Array Sizes')

Expected Output



- Bar graph showing beamforming gain for different antenna array sizes. Objective – 2

Compare Beam Width (Degrees)

Analyze the beam width corresponding to different antenna array sizes using MATLAB. Observe

how increasing the number of antenna elements results in a narrower beam, thereby improving

spatial resolution.

MATLAB Code:

```
clc;
```

```
clear;
```

```
close all;
```

```
antennas = [2 4 8 16 32]; % Number of Antenna Elements
```

```
beamwidth = [90 45 22.5 11.25 5.625]; % Beam Width (degrees)
```

figure

```
plot(antennas, beamwidth,'o-','LineWidth',2)
```

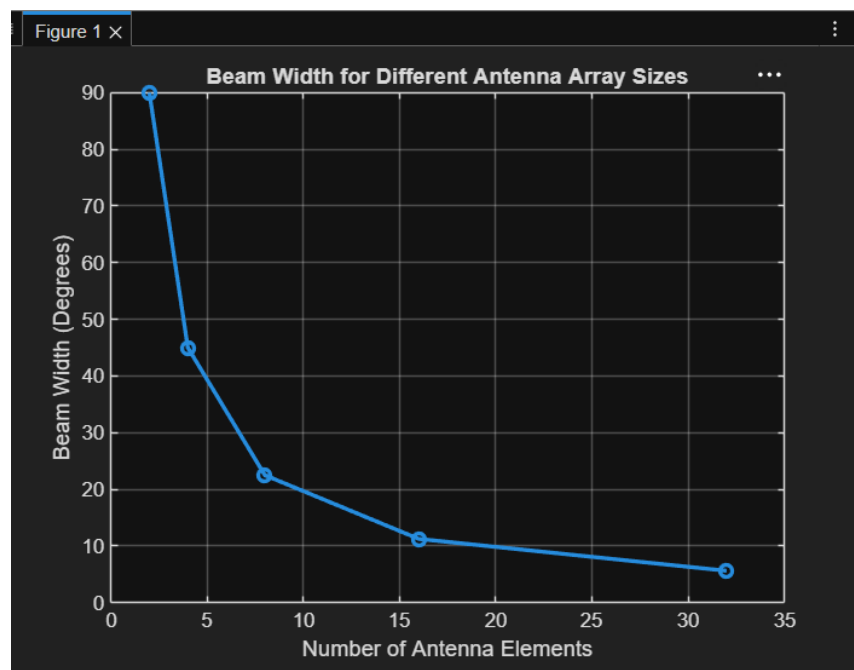
grid on

```
xlabel('Number of Antenna Elements')
```

```
ylabel('Beam Width (Degrees)')
```

```
title('Beam Width for Different Antenna Array Sizes')
```

Expected Output



- Line graph showing beam width versus antenna array size.

Objective – 3

Analyze Main Lobe Direction (Degrees)

Simulate beam steering by varying the steering angle of the antenna array using MATLAB.

Observe the change in the main lobe direction and understand how analog beamforming directs

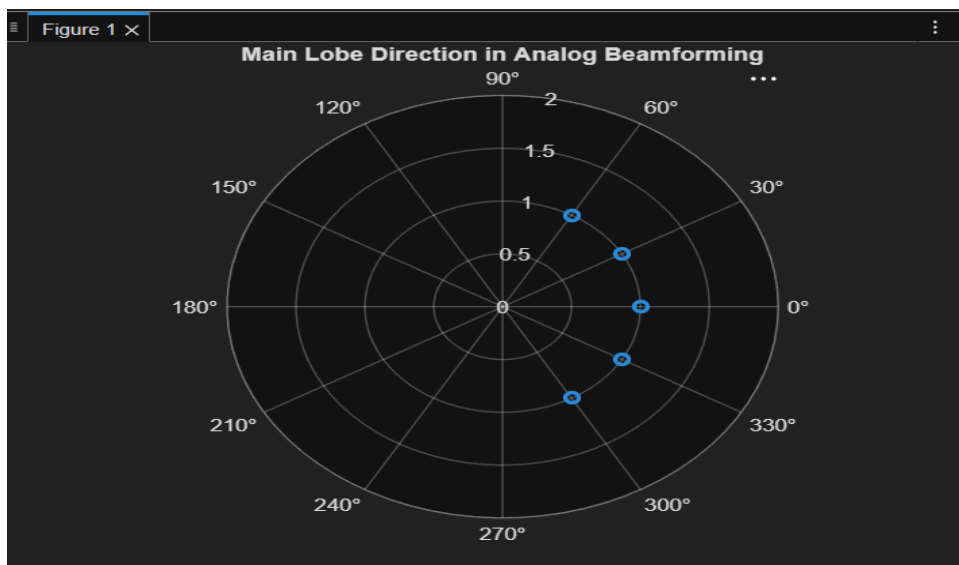
energy toward the intended user.

MATLAB Code:

```
clc;
clear;
close all;

theta = [-60 -30 0 30 60]; % Main Lobe Direction (degrees)
gain = [1 1 1 1 1]; % Constant normalized gain
figure
polarplot(deg2rad(theta), gain, 'o', 'LineWidth', 2)
title('Main Lobe Direction in Analog Beamforming')
```

Expected Output



- Polar plot or beam pattern showing the main lobe steered to different angles.

Result

The MATLAB analysis successfully demonstrated the characteristics of analog beamforming. The results showed that increasing the number of antenna elements

improves beamforming gain and reduces beam width, while phase adjustment enables the main lobe to be steered toward different directions.